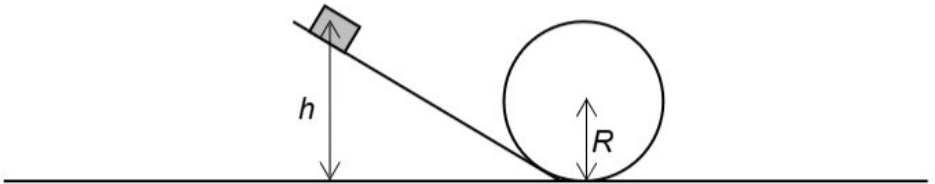
**14** A small block of mass $m$ at height $h$ is released from rest and slides down a frictionless track to enter a vertical loop of radius $R$.

What is the normal contact force exerted by the block on the track at the bottom of the loop?




**A** $mg$ **B** $mg\left(1+\frac{2h}{R}\right)$ **C** $mg\left(1+\frac{h}{R}\right)$ **D** $\frac{2mgh}{R}$